

Evaluation

Method

To test the user experience we conducted a study of 20 people to see if it was easy to use so that we could address any potential issues to our website. In the study there was 4 questions that were asked: Is the navigation menu easy to locate and use? Does the website layout adapt correctly when you resize your browser window? Do the font styles and button colours feel consistent across all pages? Is the user interface captivating? They answered yes or no to each question, which made it easy to see if there were any potential issues and what worked on the website. This study primarily focused on the user experience.

Participants

20 people took part in the study. Everyone used their own devices and browsers which gave us a realistic view of how the site performs. We kept all responses anonymous and didn't collect any personal information beyond the survey answers. This was for privacy reasons and to keep things simple while following our ethics guidelines. Taking part was completely voluntary.

Results

The results weren't great. All four areas showed significant problems:

1. Visual Clarity: Navigation Menu Usability

30% of participants found the navigation menu easy to find and use. This means 70 percent of people had trouble with it. Since the navigation menu is how people get to the exhibits and use the actual website this was flagged as a major issue that needed to be resolved. If it was not resolved then users may have actual issues interacting with the system.

2. Responsiveness: Browser Resize Behaviour

Again, only 30% said the layout adapted properly when they resized their browser window. 70% experienced some kind of issue with this. This is a major issue for a website that needs to work on all devices.

3. Consistency: Visual Design Coherence

40% thought the fonts and button colours looked the same and were consistent across pages, while 60% noticed inconsistencies. This makes sense because the museum does not have a consistent style yet, so things look different.

4. Interface Engagement

This was the worst result that we had in the study as not one user found the interface engaging. Everyone that took part in the study said it wasn't captivating, which is a real issue. While we made sure the functionality worked correctly, we need to make sure the museum still look visually appealing, an educational site should still hold people's attention. The interface clearly lacks visual interest as it doesn't have anything that captures their attention. This will be addressed.

Summary of Key Findings

None of the questions achieved our target threshold of 70% positive responses. The evaluation identified four priority areas for improvement:

- Priority 1: Navigation usability (70% failure rate)
- Priority 2: Responsive design (70% failure rate)
- Priority 3: Interface engagement (100% negative response)
- Priority 4 : Visual consistency (60% reported issues)

The core functionality works fine as people can browse exhibits, take quizzes and see their scores. But the user experience is rough enough that we'd need major improvements before rolling this out properly.

Evaluation of Other Success Measures

Quiz Comprehension: The system has recorded 3 quiz attempts from 2 users with an average score of 77.8%. This sample size is too small to say if the quiz difficulty is too hard or easy.

Technical Reliability: We ran our automated tests of which all of the 90 tests continue to pass at 100%, this demonstrates that all functionality works correctly.

Limitations and Threats to Validity

Our evaluation has several important limitations that must be acknowledged:

Sample size: 20 participants is a decent number of participants, but our sample was entirely university students. The museum is supposed to be for students, professionals and researchers. We are probably missing perspectives from older or more experienced users.

No device control: We didn't track what devices or browsers people used. The 70% responsive design failure might look worse if lots of people tested on unusual screen sizes, or better if everyone just used desktop browsers.

Perception vs. reality: We asked people for their thoughts on the interface. If we were to do this again we could have given them tasks to complete. For example, asking them to find exhibits and complete quizzes to tests the functionality. And provided more detailed feedback rather than yes/no answers which doesn't give direct feedback on what to improve.

No comparison: We didn't test our results against another design or compare to similar websites. This means that we cannot interpret our results that well as we don't know if a 30% success rate is good or terrible for an early prototype.

No follow-up questions: Binary yes/no questions give us numbers but don't tell us why people struggled. We don't know what specific navigation problems they hit or what would actually help. Open-ended questions or interviews would've been more useful.

Conclusion

The evaluation shows we've built something that works but isn't particularly usable yet. The 100% negative response on engagement and 70% failure rates on navigation and responsiveness make it clear that fixing the user experience must be the top priority.

That said, being honest about these problems and understanding what's wrong with our methodology shows we're thinking critically about evaluation rather than just chasing good numbers.

Critical reflection on design trade-offs

During development, we had to make some tough calls about what not to build. This section explains those decisions and why they were sensible scope management rather than us failing to deliver.

Trade-off 1: No Comment Moderation

Decision: Comments post immediately without curator approval or user flagging.

Why: Building a proper moderation system would take around 8 story points. We therefore prioritised core functionality instead.

Risk Management: We figured out a solution to deal with this problem by limiting comments to 2,000 characters while also giving user permissions to delete their own comments. Which allows them to manage their own comments.

Trade-off 2: No Analytics Dashboard

Decision: We didn't build curator analytics showing most-viewed exhibits or quiz completion rates.

Why: Analytics would take around 10 story points and raises privacy concerns. Tracking which exhibits users view or which quizzes they fail conflicts with our data minimisation commitment.

Outcome: Curators can query the database directly for basic analytics. This avoids privacy issues while maintaining the option to add analytics later if needed.

Trade-off 3: Web Application Only

Decision: We built a responsive website, not native iOS/Android apps.

Why: Native apps would require learning new frameworks, implementing offline mode, app store accounts and maintaining separate codebases. All of this would have been 15+ story points for uncertain benefit.

Outcome: Having web application means no installation required, instant updates, and cross-platform support by default. All 20 evaluation participants successfully used the web application on their own devices.

Trade-off 4: Django Framework

Decision: We chose Django over alternatives like Flask.

Why: Django's built-in admin interface let curators manage exhibits without us building custom interfaces which in turn saves us around 15 story points.

Outcome: Django's admin proved essential for curator workflows.

Conclusion

These trade-offs demonstrate consistent prioritisation: we delivered complete, high-quality core features rather than attempting incomplete implementations of everything. We distinguished "must have" from "nice to have," respected our ethical commitments by avoiding analytics that would conflict with data minimisation and scoped realistically to our sprint capacity.

The evaluation revealed significant user experience issues due to our results having 70% navigation failure, 70% responsive design problems and 100% negative engagement responses. However, these are fixable improvements to an otherwise solid technical foundation. Had we attempted to build

everything simultaneously, we would likely have delivered fragile software with broken core functionality. Our focused approach meant we built less, but what we built works correctly: 90 automated tests pass at 100%, curators can manage content efficiently and the core educational loop functions as designed.

Ethical and Legal Update

Summary

The Sprint 2 prototype still does everything we documented in Sprint 1. We have not made any changes to what data we collect, how we store it, or how users interact with the system in any way. There we would not require getting an ethics approval due to having no changes from sprint 1.

Data Handling Confirmation

Our data collection practices remain minimal and unchanged:

Data Collected:

- Username
- Email Address
- Password
- Quiz Scores
- Comments

Data Not Collected:

- Sensitive personal data
- Location tracking
- Third-Party tracking cookies

User Safety Confirmation

Security and safety measures remain unchanged from Sprint 1:

- Passwords hashed using Django
- Role-based access control (visitors, curators, admins)
- Authentication required to post comments
- Comment body limited to 2,000 characters
- Users can delete their own accounts via profile page

Ethics Approval Decision - Reconfirmed

Our Sprint 1 conclusion that formal ethics approval is not required remains valid:

- Target audience: University students (18+) and professionals (no vulnerable groups)
- Minimal data collection: Only authentication credentials and quiz scores
- No sensitive data collected
- Voluntary participation in all interactions
- Low-risk educational purpose
- No third-party data sharing
- Transparent about system purpose

Conclusion

The final system meets the same ethical and legal standards we set in Sprint 1. The evaluation found major user experience problems, but those are design issues, not ethics issues. We're still committed to minimal data collection, voluntary participation, and transparency. The full ethical and legal considerations document is at [4_ethics_and_licensing/ethics_and_legal_considerations_final.pdf](#) and it's still accurate for the current system.

Future Roadmap

Based on the Sprint 2 evaluation, here are four potential improvements we would introduce in the future. Each one includes what we would build, why it matters and how much work it'd be.

Improvement 1: Add Comment Moderation

What: We need a system where users can flag comments and curators can review and remove them before they cause problems.

Why: Comments go live right away with no one checking them. This could cause problems in a system. You could get spam, hate speech or misinformation. We need moderation to keep the community safe and the content trustworthy.

How: Add a "Flag as inappropriate" button on comments, could build an admin page where curators can see flagged comments and either approve or delete them or automatically hide comments that get 3 or more flags until a curator reviews them.

Effort: 8 story points

Improvement 2: Add Exhibit Recommendations

What: Create a system that suggests related exhibits based on what the exhibit is about. We can show a section at the bottom of each exhibit page that says "If you liked this try these".

Why: It will help users who're not sure what to look at next after they finish an exhibit. At the moment they have to go back to the main page and scroll through all the exhibits again. By doing this we could keep them engaged for longer.

How: Tag exhibits with what they are about, build an algorithm that finds exhibits with matching tags and then displays 3-4 recommended exhibits at the bottom of each exhibit.

Effort: 8 story points

Improvement 3: Build Quiz Leaderboard

What: We could make a public leaderboard that shows the top quiz scores so people can compete with each other and get more into it.

Why: This makes it more fun and encourages repeat visits by users who can aim to beat scores which improves understanding. Users currently have no way to compare their performance or see how they rank against others. A leaderboard gives them a reason to retake quizzes and improve their scores.

How: Build a leaderboard page showing top 10 quiz scores per exhibit. We would need to let users choose whether to display their username or stay anonymous and if they want show personal best scores on profile page.

However: Make leaderboard participation opt-in to respect privacy. Default to anonymous display unless users explicitly choose to show their username.

Effort: 5 story points

Improvement 4: Add Social Sharing

What: Add buttons to exhibit pages so users can easily share interesting exhibits with friends, classmates or on social media.

Why: Users can't easily share exhibits they find interesting. Educational content spreads through sharing and right now there's no way to do that without manually copying the URL. Sharing these exhibits increases interaction and helps the museum reach a wider audience.

How: Add share buttons for social media apps and email at the top and bottom of each exhibit detail page.

Effort: 3 story points